

3. HOW TO APPLY SHORT HIGH INTENSITY INTERMITTENT INTERVALS WITH PROGRESSIVE OVERLOAD

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Knowledgeiswatt

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SCIENTIFIC BACKGROUND:

Short high intensity intermittent intervals are a HIT format in which the work duration is 15sec-1 min and the active-recovery ratio is 1:1 or higher (for example: 3:2 or 2:1).

In previous KIW Research Notes, we saw that **short high intensity intermittent intervals promotes superior aerobic physiological and performance adaptations** (VO₂max, power at thresholds, 20min TT) **when compared to long high intensity interval**. This has been shown both in trained amateur (1) and professional (2) road cyclists.

HIGH INTENSITY TRAINING (HIT) IN TRAINED (NON-PRO) CYCLISTS: SHORT OR LONG INTERVALS?

20 trained but non-pro cyclists (~66 VO₂max) were divided into two groups which for 10 weeks performed the same training program (10–12 hrs per week). The only difference was in the HIT sessions they performed 2x week (the workout were performed trying to have the highest average power possible over all the intervals):

SI (Short Interval Training)

3x 13 x 30s on-15s off
3' rec.



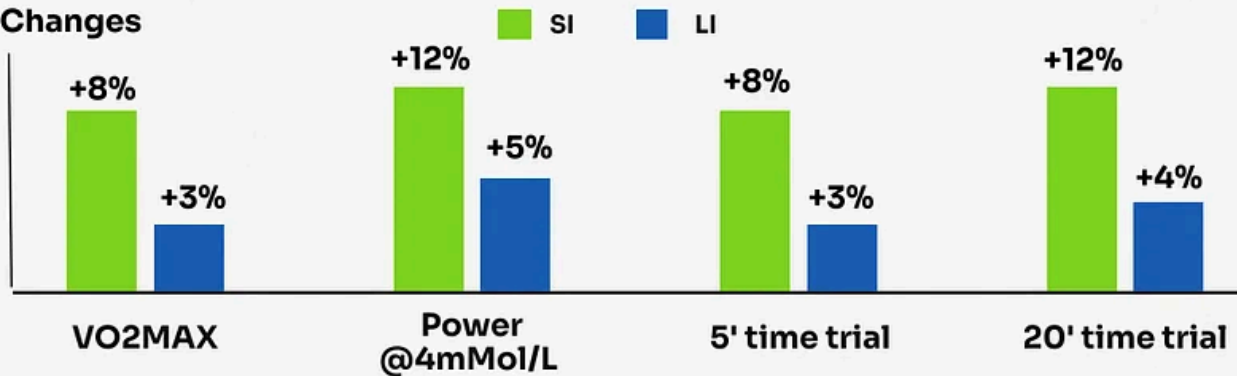
LI (Long interval Training)

4 x 5' on
2.5' rec.



RESULTS

% Changes



Short HIT Intervals produces superior training adaptations compared with Long HIT Intervals in well-trained but non-pro cyclists

Rønnestad et al. 2015



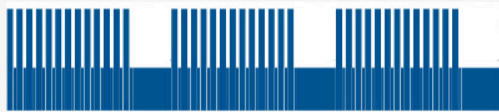
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SHORT VS LONG HIGH INTENSITY INTERVALS IN PRO CYCLISTS

18 pro cyclists were divided into two groups and performed 3 times per week for 3 weeks two different interval training sessions at the maximum sustainable intensity (max effort).

SI (Short Interval Training)

3x 13 x 30s on-15s off
3' rec.

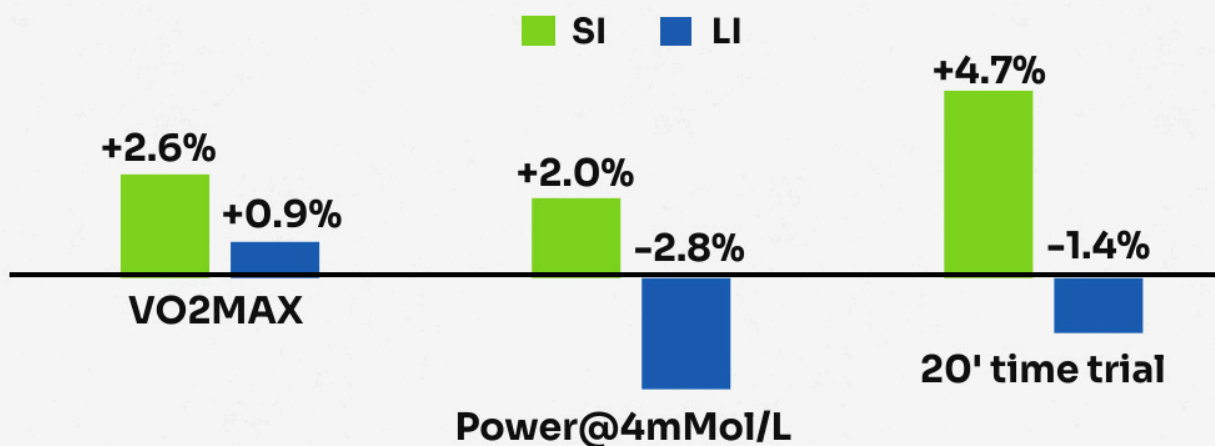


LI (Long interval Training)

4 x 5' on
2.5' rec.



RESULTS



Short high intensity intervals produce superior aerobic training adaptations compared to Long HIT Intervals in professional cyclists

Rønnestad et al. 2020



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Specifically, in both the two cited studies, the workouts compared were:

- short high intensity intermittent intervals:

3x13' alternating 30" 'on'/15" 'off' with 3' recovery at high zone 1 or low zone 2 (using a 5/7 zoned model).

- long intervals:

4x5' 'on' with 2.5' recovery at high zone 1 or low zone 2 (using a 5/7 zoned model).

If you make some calculations, you can see as both short and long HIT workouts accumulated a similar time at high intensity (sum of the 'on' phases): 19min30s vs 20 min. In both HIT's formats, the 'on' phases were performed at the maximum sustainable effort to obtain the highest average power throughout all the intervals. Interestingly, despite the similar rate of

perceived exertion, the average power during the 'on' phases was higher in the short intermittent vs long intervals (~130% vs ~115% of second threshold). So, **it seems like the capacity to work at a higher intensity despite a similar rate of perceived exertion and total time spent at a high intensity power is what make short intervals superior in promoting aerobic adaptations**

This is line with a very recent couple of studies which showed that:

1. **Working at a higher fraction of maximal oxygen consumption (VO₂max) is the key factor to get the most from HIT sessions for what concern aerobic physiological and performance gains; (3)**
2. **The short high intensity intermittent intervals are the HIT format which permit to optimize time spent at high fraction of VO₂max when compared to other HIT formats, such as steady long intervals and under-over intervals (alternating phases just above FTP to phases to just below FTP); (4)**

ON AVERAGE, WHICH HIT FORMAT IS THE BEST TO OPTIMIZE AEROBIC ADAPTATIONS?

19 trained amateur cyclists (~65 VO₂max) performed on 3 different days, 3 different high intensity training (HIT) sessions in the lab. In all the 3 formats, the recovery between the 5x8min efforts was 3min at 35% of PO₄₀min. PO₄₀min is the personal best mean power output for 40min.

INTERMITTENT SHORT INTERVALS (ISI)

5x8min alternating 30-s at 118% of PO₄₀min and 15-s at 60% of PO₄₀min

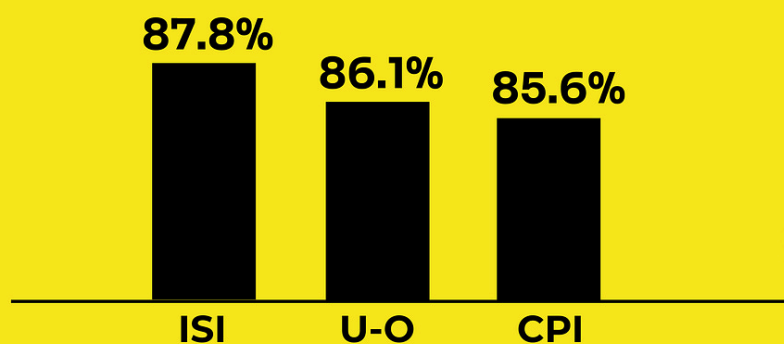
UNDER-OVER INTERVALS (U-O)

5x8min alternating 60-s at 110% and 90% of PO₄₀min

8' CONSTANT PACE INTERVALS (CPI)

5x8min constant at 100% of PO₄₀min

AVERAGE OXYGEN CONSUMPTION (AS % OF VO₂MAX) ATTAINED DURING THE DIFFERENT FORMATS



On average, intermittent short intervals are the HIT format that permits to accumulate more time at high fraction of VO₂max and so should be the most effective to induce positive aerobic adaptation

Urianstad et al. 2024



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So, the 'superpower' of short high intensity intermittent intervals seems to be the following:

while the specific number of series, intervals' length and intensity of the 'on' phases, can vary between different prescribed workouts, when matching rate of perceived exertion and total time spent at high intensity power output, they permit to accumulate more time at higher fraction of VO₂max and so they stimulate more aerobic physiological and performance gains when compared to other HIT formats!

The high intensity intermittent short intervals' workout used by the above cited studies (3x13' alternating 30" 'on'/15" 'off' with 3' recovery) could give a strong performance boost in the short term. However, when starting to put this HIT format in a training plan, I suggest to do not copy and past it: it is quite hard! I would rather start with a lighter version using less repetitions, less series and/or a lower on-off ratio, and then **applying a progressive overload through the weeks manipulating (increasing) these variables:**

1. **Number of series**
2. **Number of repetitions**
3. **On-off work ratio.**

This can be important to avoid a 'from 0 to 100' strategy (from no short intermittent high intensity intervals to its 'hardest' version) and **to maintain space to increase the load within this HIT format through the weeks, which might be crucial to maintain responsiveness to this stimulus and to avoid stagnation in performance gains.**

Here below I would show you a practical example on how to do it.

A PRACTICAL EXAMPLE ON HOW TO APPLY THE SHORT HIGH INTENSITY INTERMITTENT INTERVALS WITH PROGRESSIVE OVERLOAD:

- **The Athlete:** I think this type of progression is not influenced by the level of the athlete considered. Given that it is based on relative intensity (rate of perceived exertion and/or % of thresholds/FTP/Critical power), everyone should be able to perform it (if the rest of the training plan is well structured and balanced).
- **The Goal Race:** I think this progression can be put in many different scenarios where aerobic development is the focus of the training mesocycle. Not just for certain athletes or certain races.
- **The Periodization:**
 - Macro-periodization: I wouldn't put it too early in the annual periodization. I would insert it after at least 6—8 weeks of 'base' training with focus on low and medium intensity.
 - Micro-periodization: I wouldn't do more than 2 sessions per week. I suggest each session should be followed and preceded by a day in which medium/high intensity is not the focus (to maintain a good alternation between load and recovery, especially from an autonomic point of view).

*An exception could be HIT block periodisation for time-crunched cyclists, where even 4/5 consecutive days of this format are performed (followed by at least 4/5 days of recovery/taper).

- **The Progressive overload:**

1. Start with 2 sets of 8 minutes with 1:1 work-recovery ratio.
2. Increase from 2 to 3 sets.
3. Increase the length of each sets: from 8 to 12 min (I would not go over 13min).
4. Increase the work recovery ratio from 1:1 to 3:2.

When doing this, in the first new session(s) you can decrease the length of each sets (from 12 to 10min).

5. (Re)-Increase the length of each sets (from 10 to 12min), while maintaining the 3:2 work-recovery ratio.
6. Increase the work-recovery ration from 3:2 to 2:1.

When doing this. in the first new session(s) you can decrease the length of each sets (from 12 to 10min).

7. (Re)-Increase the length of each sets (from 10 to 12min), while maintaining the 2:1 work-recovery ratio.

	Session 1	Session 2
Week 1	2x8 min 30"-30"	3x8min 30"-30"
Week 2	3x10min 30"-30"	3x12min 30"-30"
Week 3	3x12min 30"-30"	3x10min 30"-20"
Week 4	3x10min 30"-20"	3x12min 30"-20"
Week 5	3x10min 40"-20"	3x10min 40"-20"
Week 6	3x11min 40"-20"	3x12min 40"-20"

What should be the intensity of the 'on' phases?

Start with 120-130% of FTP/CP/second threshold, but it can be higher (140-150%) for athletes with a very big anaerobic reserve (W'). After seeing the values recorded in the first sessions (power, heart rate, perceived exertion) I manipulate the work intensity based on the specific moment of the season and the session's goal. When I want a 'breakthrough' session, I prescribe to go for the higher average power attainable for all the intervals, with also sometimes suggesting a challenging specific target power output that I think is doable based on the previous athlete's data. On the other hand, when I want a solid high intensity aerobic session but without going too deep (to save physical and mental energy for the medium-long term), I suggest to go for 115-120% of FTP/CP/second threshold.

Thanks for reading! Share with your Team!

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